

GSM-based Dot-matrix Display



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Nowadays, LED message displays are becoming very popular for transmitting information to large groups of people quickly. These can be used indoors

or outdoors, in places such as railway platforms, public transport systems, traffic signals, highway signboards, banks, public offices, hotels, training institutes, nightclubs and shops. Compared to LEDs, liquid-crystal displays (LCDs) are easy to interface with a mi-

crocontroller is a GSM module with a dedicated SIM card inserted and a microcontroller unit (MCU1). When a new message is received, MCU1 receives the corresponding frame from the GSM module. MCU1 then processes the data received and extracts the message from the complete frame. Once a new message is extracted, the GSM section informs the display section and sends the message via serial interface.

Display section. This section comprises a 7×40 dot-matrix display and a microcontroller unit (MCU2). MCU2 receives the message from MCU1 on serial interface. This message is stored in an array. The message is processed and corresponding serial data are sent to the LED drivers (shift registers) that are connected in the daisy chain.

Fig. 3 shows the circuit of GSM-based dot-matrix display. It comprises two P89V51RD2 microcontrollers (IC9 and IC10), regulator 78S05 (IC12), shift registers (IC1 through IC8), MAX232 driver (IC11), MOSFETs (T1 through T5), 5×7 dot matrix modules (DIS1 through DIS8), GSM module SIM300 and a few discrete components.

The 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 3A. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C12 and regulated by IC12. Capacitor C13 bypasses the ripples present in the regulated supply. The generated 5V is supplied to the complete circuit. LED1 acts as the power indicator and R8 limits the current through LED1.

The microcontroller used in the GSM and display sections is P89V51RD2, which is an 8-bit microcontroller with 64kB Flash and 1024 bytes of data RAM. The P89V51RD2 has three 16-bit timers/counters, programmable watchdog timer, eight interrupt sources with four priority



Fig. 1: Author's prototype

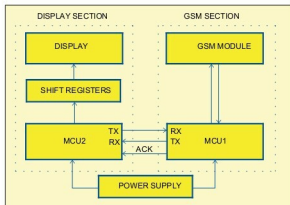


Fig. 2: Block diagram of GSM-based moving message display

crocontroller for displaying information as these have many built-in functions, but these can't be observed from a distance, and large-size LCDs are very costly.

Presented here is a GSM-based, eight-digit dot matrix display that can show an eight-digit message sent via any mobile phone, from any place around the world, to the mobile number

of the SIM in the GSM module of the display. Size of the display is 7×40, which is made using 5×7 dot-matrix LED modules.

Circuit and working

Fig. 2 shows the block diagram for GSM-based moving message display. For simplicity, the complete circuit is implemented into two parts: GSM section and display section.

GSM section. The GSM section

Test Points for Fig. 1

Test point	Details
TP0	0V, GND
TP1	5 V
TP2	Serial data
TP3	Square pulses
TP4	Square pulses
TP5	Square pulses
TP6	Square pulses
TP7	Square pulses